

# Daniel Traian Pele


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Senior Researcher I (CS I), Institute for Economic Forecasting (IPE), Romanian Academy
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## ABOUT ME

More recently, I study how large language models and time-series foundation models can be used to forecast risk, partly in joint work with my PhD students and the partners of the MSCA Doctoral Network on Digital Finance.

## WORK EXPERIENCE

2022 – CURRENT Bucharest, Romania

**Senior Researcher I (CS I)**

**Institute for Economic Forecasting (IPE), Romanian Academy**

Head of the Department of Computer-Assisted Economic Research.

2020 – CURRENT Bucharest, Romania

**Professor**

**Bucharest University of Economic Studies (ASE), Department of Statistics and Econometrics**

Courses: Statistics, Econometrics, Time Series, Financial Markets Statistics, Advanced Time-Series Modelling, Financial Markets Modelling; supervision of bachelor and master theses.

2019 – CURRENT Bucharest, Romania

**PhD supervisor**

**Doctoral School of Cybernetics and Economic Statistics, ASE**

2018 – 2019 Berlin, Germany

**Researcher, IRTG 1792 "High Dimensional Nonstationary Time Series"**

**Humboldt-Universität zu Berlin, with Prof. Wolfgang Karl Härdle**

2016 – 2020

**Statistical expert**

**European Investment Bank, European Commission, National School of Political and Administrative Studies**

Sampling methodology for the ex-ante control of public procurement (EIB); minimum-wage setting mechanism for Romania (European Commission, INCSMPS); cost standards for local public services (SIPOCA 9).

2014 Berlin, Germany

**Visiting researcher**

**Research Data Center, Humboldt-Universität zu Berlin, with Prof. Wolfgang Karl Härdle (Quantlet platform)**

Development of the Quantlet platform in SAS for Statistics of Financial Markets.

2014 – 2024 Bucharest, Romania

**Statistical expert**

**The World Bank: tax framework, public procurement, disaster-risk financing, governance scorecards**

Tax framework under Romania's Recovery and Resilience Plan; public procurement strategy; disaster-risk financing and the Bucharest seismic-risk tool; Country Climate and Development Report for Azerbaijan; Moldova Governance Scorecard; Romania Strategy Unit; sampling methodology for the Human Capital Operational Programme.

2014 – 2015

**Statistical expert, Delivery Unit**

Prime Minister's Chancellery, Romania (World Bank project)

Statistical analysis, monitoring and evaluation, government dashboard, risk model for VAT refunds; analyses of public procurement, youth employment, tax administration and energy.

2007 – 2021 Bucharest, Romania

**Statistician**

Ipsos Interactive Services, Bucharest: statistical modelling

2007 Bucharest, Romania

**Senior adviser**

National Institute of Statistics, Bucharest: statistical analysis and sampling

2004 – 2005 Bucharest, Romania

**Risk analyst**

Credisson International, Bucharest: scoring models

2001 – 2020 Bucharest, Romania

**Teaching assistant (2001), assistant (2003), lecturer (2007), associate professor (2013)**

ASE, Department of Statistics and Econometrics

## EDUCATION AND TRAINING

2019 Bucharest, Romania

**Habilitation, Cybernetics and Statistics**

ASE. Thesis: Statistical methods for financial markets

2011 Reading, United Kingdom

**Postdoctoral fellowship**

ICMA Centre, University of Reading, with Emese Lazar: uncertainty in capital markets via information entropy

2010 – 2013 Bucharest, Romania

**Postdoctoral researcher**

ASE: predictability of financial crises

Topic: predictability of financial crises, with applications to the stock market. Advisers: Nicolae Dardac, Gabriela Anghelache.

2007 Bucharest, Romania

**PhD in Economics (Cybernetics and Statistics)**

ASE. Advisor: Vergil Voineagu

Thesis: Statistical methods for the analysis of the stock market.

2002 Bucharest, Romania

**MSc in Statistics (Stochastic Processes and Theoretical Statistics)**

University of Bucharest, Faculty of Mathematics. Advisor: Monica Dumitrescu

Dissertation: Repeated surveys through linear time-series models.

2000 Bucharest, Romania

**BSc in Mathematics (Advanced Studies)**

University of Bucharest, Faculty of Mathematics. Advisor: Anton Ștefănescu

Thesis: Competitive equilibrium in the Arrow–Debreu model.

1992 – 1996 Corabia, Romania

**High school**

Alexandru Ioan Cuza National College (then Theoretical High School), Corabia, Olt County

## LANGUAGE SKILLS

**Mother tongue:** Romanian

**Other languages:**

| | Listening | Reading | Spoken interaction | Spoken production | Writing |
|---|---|---|---|---|---|
| **English** | B2–C1: certified overall level (PROSPER-ASE certificate no. 8472, 30.06.2022) | | | | |
| **French** | B2 | B2 | B2 | B2 | B2 |

Levels: A1 and A2 basic user; B1 and B2 independent user; C1 and C2 proficient user (Common European Framework of Reference for Languages).

## DIGITAL SKILLS

Statistical and econometric software: SAS, R, MATLAB, SPSS, EViews, Stata. Programming and data science: Python (pandas, statsmodels, scikit-learn, PyTorch), Jupyter and Google Colab. Scientific writing and reproducible research: LaTeX and Beamer, Git and GitHub, Quantlet. Microsoft Office.

Self-assessment (Europass digital competence grid): proficient user in information processing, communication, content creation, safety and problem solving.

## ORGANISATIONAL SKILLS

- Founded the SAS Centre of Excellence at ASE Bucharest.
- Organised the AIDA Conference 2026 (AI and Digital Assets in Energy and Finance, Bucharest) and the workshop “AI, Digital Assets and the Future of Energy Finance” (2024); minitrack chair at ICBE.
- Project director for ASE in the MSCA Doctoral Network DIGITAL and co-leader of Work Package 4.

## PROJECTS

2024 – CURRENT Bucharest, Romania

**DIGITAL: MSCA Industrial Doctoral Network on Digital Finance**

Horizon Europe, No. 101119635

Project director for ASE and co-leader of Work Package 4, "Driving digital innovation with Blockchain applications"; industrial doctoral training in AI for finance.

2023 – 2026

**IDA: Institute for Digital Assets**

PNRR, contract CN760046/23.05.2023

Deputy Director. Research: foundation models and LLMs for risk forecasting (LLM-VaR, LLM-ES), machine learning for digital assets (early warning for “zombie” assets, liquidity forecasting).

2023 – 2026

**AI4EFin: Artificial Intelligence for Energy Finance**

PNRR, contract CN760048/23.05.2023

Researcher; led the research line on forecasting, explainable AI and statistical calibration and coordinated the PhD students in the team: energy-price forecasting, early warning for price spikes, conformal calibration.

2020 – 2024

**COST Action CA19130: Fintech and AI in Finance**

Management Committee member for Romania; trainer in the Action’s summer schools.

2019 – 2021

**FIN-TECH: Horizon 2020**

Horizon 2020, No. 825215

Financial supervision and technology compliance training programme: an EU-wide risk-management platform.

2018 – 2019 Berlin, Germany

**IRTG 1792: High Dimensional Nonstationary Time Series**

Humboldt-Universität zu Berlin

Researcher on the genus–differentia classification of cryptocurrencies, with Wolfgang Karl Härdle.

2021

**Unified methodology for damage assessment in emergencies and disasters**

World Bank and the General Inspectorate for Emergency Situations, contract no. 111823/20.08.2021

2012 – 2013

**Pre-financing rate of the EU structural instruments**

ACIS, contract no. 3/23/57/26.09.2012: econometric model of the impact of pre-financing on the absorption rate

2011 – 2012

**Institutional capacity of the National Forecasting Commission (SMIS 27153)**

Studies on the contribution of SMEs to economic growth and on the private sector in research and innovation: model of the impact of R&D spending on growth

2011 – 2012 Bucharest, Romania

**Professionalism in the county statistics offices (SMIS 15966)**

ASE with the National Institute of Statistics

2009 – 2011

**CNCSIS IDEI research grants ID_1820 and ID_183**

Team member

2008 – 2011

**Quality in education (CALE), POSDRU**

Econometric model forecasting labour demand in construction under the macroeconomic and crisis context

2006

**CEEX research project**

Team member

2002 – 2006

**CERES research projects**

Team member

## NETWORKS AND MEMBERSHIPS

- Associate Editor, *Digital Finance* (Springer)
- Member, Romanian National Statistical Council (2023–)
- Board member for Romania, European Courses in Advanced Statistics (ECAS, 2020–)
- Management Committee member, COST Action CA19130 (2020–2024)
- Minitrack chair, International Conference on Business Excellence (ICBE 2023, “Applying ML and AI to Finance, Business and Society”; ICBE 2025, IDA minitrack)

## HONOURS AND AWARDS

- Editor of Distinction Award (Author Service Award 2026), Springer Nature, for editorial work at Digital Finance
- Best Paper Award, International Conference on Business Excellence (ICBE) 2026, for “A Multimodal Vision-Language Framework for Financial Anomaly Detection”, with S.-L. Jheng, R. Tak and Ș. Găman
- Honorary Citizen of Gura Padinii commune, Olt County (2024)
- Best Paper Award, International Conference on Business Excellence (ICBE) 2019 (Journal of Economic Forecasting Award), with W. K. Härdle and N. Wesselhöfft
- "Bologna Professor" diploma (2010)

- Diplomas of excellence, Romanian Statistical Society (2009, 2010, 2012)

## PUBLICATIONS

98 publications, of which 46 journal articles; full list with links and BibTeX: https://danpele.github.io/publications.html